

Amazon Sales Data Analytics Project

End-to-End ETL, Database, and BI Implementation

**A comprehensive review of 121K+ transactions resulting in ₹78.59M
in Revenue.**

Focusing on: Data Engineering, Relational Database Modeling, and
Actionable Business Intelligence.

1. Methodology: How This Project Was Built

This project was executed using a modern data stack to transform raw, messy e-commerce exports into a fully interactive business intelligence dashboard. The workflow was divided into three core technical phases.

1.1 Phase 1: Data Extraction & Cleaning (Python)

The raw Kaggle dataset contained over 128,000 rows with inconsistent naming, missing values, and problematic whitespace. Using Python (Pandas) in a PyCharm environment, I engineered an ETL script to:

- Normalize columns using snake_casing for SQL compatibility.
- Strip hidden whitespaces and drop strictly empty financial rows.
- Interpolate null demographic data to preserve transaction volume metrics.

2. Data Engineering Proof of Execution

Below is the programmatic cleaning phase where raw data was standardized. Python successfully processed and retained 121,180 verified rows for database ingestion.

```
4 # -----
5 # 1. READ AND CLEAN THE DATA
6 # -----
7 df = pd.read_csv(r"C:\datasets\archive (3)\Amazon Sale Report.csv", low_memory=False)
8 print(f"Extraction Phase: Python sees exactly {len(df)} rows of raw data.")
9
10 # Snake casing
11 df.columns = df.columns.str.lower()
12 df.columns = df.columns.str.replace(' ', '_')
13 df.columns = df.columns.str.replace('-', '_')
14
15 # Deleting unnecessary columns
16 df = df.drop(columns=['unnamed:_22', 'order_id', "index", "asin"])
17
18 # Convert date into datetime datatype
19 df["date"] = df["date"].astype("datetime64[ns]")
20
21 # Remove whitespace
22 whitespacecolumns = ['status', 'fulfilment', 'sales_channel_',
23                      'ship_service_level', 'style', 'sku', 'category', 'size',
24                      'courier_status', 'qty', 'currency', 'ship_country', 'b2b']
25
26 for col in whitespacecolumns:
27     df[col] = df[col].astype(str).str.strip()
```

mMiscProject > amazon_salesa.ipynb 2:1 LF

2.1 High-Speed Database Ingestion

Instead of manual CSV uploads, I wrote a strict SQL schema (`CREATE TABLE`) and utilized the `psycopg2` library to establish a direct connection to PostgreSQL, executing a high-speed bulk upload.

```
amazon_sales.ipynb  amazon_salesa.ipynb x
| ↑ ↓ | ↵ ⏏
121 # Save everything
122 conn.commit()
123 print("🚀 Upload complete! All rows inserted successfully.")
124
125 except Exception as e:
126     print("❌ An error occurred:", e)
127
128 finally:
129     if 'conn' in locals() and conn:
130         cursor.close()
131         conn.close()
132     print("🔒 Database connection closed.")

[1]
```

Extraction Phase: Python sees exactly 128975 rows of raw data.
Cleaning Phase Complete: Preparing to upload 121180 clean rows.
✅ Connection succeeded to psql database!
🚀 Building the table structure...
✅ Table created successfully!
🚀 Uploading data...
🚀 Upload complete! All rows inserted successfully.
🔒 Database connection closed.

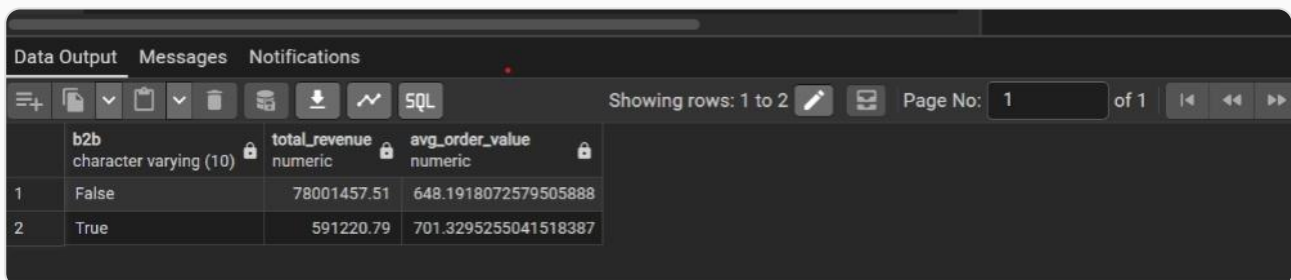
3. Phase 2: Relational Database Modeling (PostgreSQL)

Once the data was loaded into PostgreSQL, I utilized it as a robust data warehouse. This allowed me to run complex aggregate queries to discover insights before ever opening a visualization tool.

```
-- 1. The B2B vs. B2C Check
SELECT
    b2b,
    SUM(amount) AS total_revenue,
    AVG(amount) AS avg_order_value
FROM amazon_sales
GROUP BY b2b;
```

3.1 SQL Query Results

By grouping the data by the B2B flag, the PostgreSQL query revealed a critical insight: B2B customers have a noticeably higher Average Order Value.

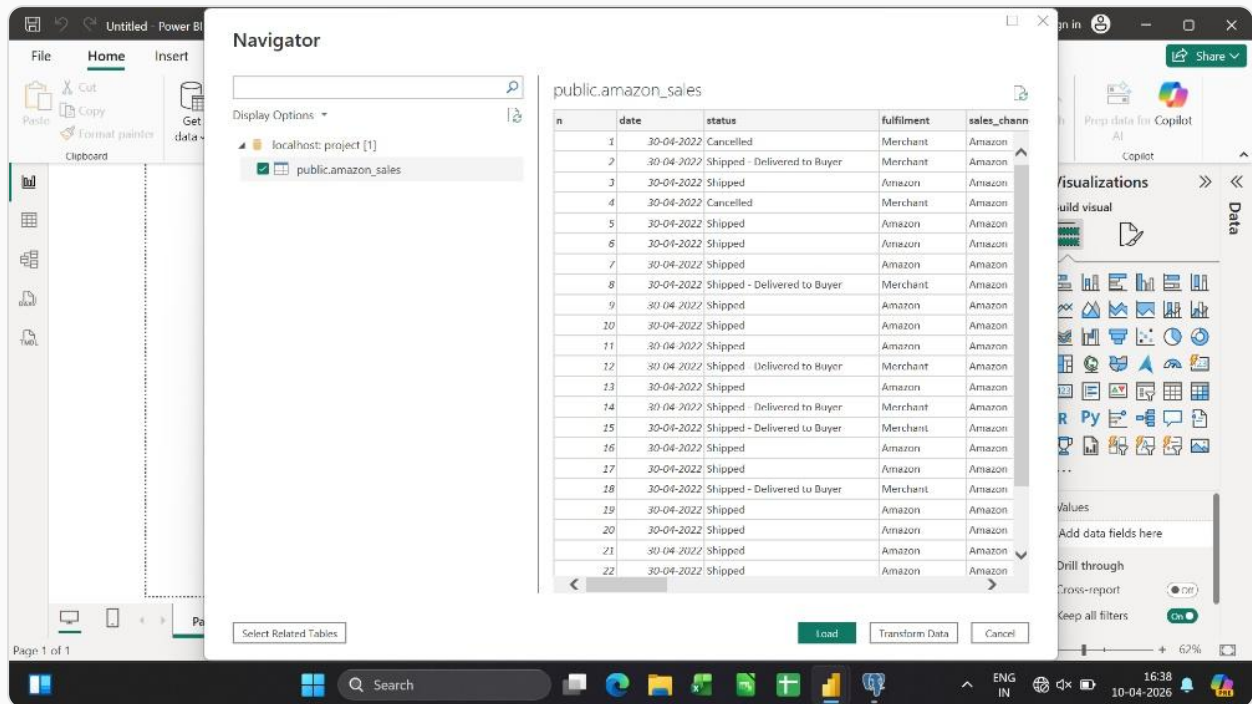


The screenshot shows a PostgreSQL query results window with the following data:

	b2b character varying (10)	total_revenue numeric	avg_order_value numeric
1	False	78001457.51	648.1918072579505888
2	True	591220.79	701.3295255041518387

4. Phase 3: Business Intelligence (Power BI)

The final phase was surfacing these insights for executive review. I connected Power BI directly to the PostgreSQL local instance to ensure a live, scalable data model.



4.1 Dashboard Architecture

The Executive Summary dashboard was built to highlight macroeconomic trends, category performance, and the massive B2C volume reliance. The clean, blue-themed UI allows managers to instantly digest the health of the business.

5. Executive Dashboard Review

The finalized dashboard synthesizes all cleaned database metrics into an interactive report.



6. Key Findings & Conclusion

Based on the end-to-end data processing and visualization, several key business realities were identified.

6.1 Core Findings

- **Macro Performance:** The dataset yielded a total validated revenue of **₹78.59 Million** across **121.18K** successful orders. The overall Average Order Value (AOV) sits at ₹648.56.
- **The April Anomaly:** The temporal trend chart indicates a massive, isolated spike in revenue during April (nearing ₹30M), followed by a stagnant decline through May and June.
- **Category Dominance:** "Sets" and "Kurtas" are the undeniable lifeblood of the product catalog, generating roughly ₹39M and ₹21M respectively, completely dwarfing categories like Ethnic Dress and Blouses.
- **The B2B Margin:** While standard retail (B2C) accounts for a massive 99.25% of the transaction volume, B2B transactions command a higher AOV (₹701 vs. ₹648).

6.2 Project Conclusion

The business operates a highly successful, high-volume retail engine, primarily driven by two apparel categories. However, the reliance on single-item B2C purchases and the sharp drop in post-April momentum indicates a vulnerability in customer retention and cart expansion. Transitioning focus from pure customer acquisition to Average Order Value optimization is required for sustained growth.

7. Strategic Suggestions & Recommendations (Part 1)

To capitalize on the findings from this data model, the following operational and marketing adjustments are recommended.

7.1 Retention Strategy for the "April Drop-Off"

The massive surge in April volume indicates a highly successful marketing campaign or seasonal event. However, the immediate decline in May and June proves the business acquired "one-time buyers" rather than loyal customers.

- **Action:** Implement a 30-day automated remarketing campaign for new buyers.
- **Action:** Audit the April promotional codes to see which discounts brought in the highest volume, and selectively re-run modified versions of those promotions in late June to revive the momentum.

7.2 Maximizing the "Set" and "Kurta" Categories

Since these two categories generate nearly ₹60M combined, they are the anchor products of the brand. Bottom-tier products (Blouses, Bottoms) are barely registering on the revenue charts.

- **Action:** Introduce "Complete the Look" bundling. When a customer adds a Kurta to their cart, aggressively offer a heavily discounted "Bottom" to go with it. This clears out dead-weight inventory while simultaneously raising the Average Order Value.
- **Action:** Reallocate ad spend away from underperforming categories and double down on the high-converting Set/Kurta demographics.

8. Strategic Suggestions & Recommendations (Part 2)

8.3 Expanding the B2B Footprint

The SQL analysis conclusively proved that B2B orders generate more revenue per transaction (₹701 vs ₹648). Yet, they currently make up less than 1% of the total volume.

- **Action:** Create a dedicated B2B sales portal or corporate gifting catalog.
- **Action:** Offer tiered volume discounts to corporate buyers (e.g., "Buy 50 Sets for corporate events, save 15%"). Because their baseline spend is already higher, incentivizing bulk B2B purchases is the fastest way to scale top-line revenue without drastically increasing shipping overhead.

8.4 Fulfillment and Cancellation Optimization

While the dashboard shows massive revenue, backend logistics must protect that cash flow.

- **Action:** Conduct a strict audit of third-party merchant fulfillment times. If merchants are failing to ship on time, leading to cancellations, shift the highest-selling SKUs (Sets and Kurtas) exclusively to Amazon Fulfillment (FBA) to guarantee delivery and protect the brand's reputation.

9. Final Summary & Future Scope

9.1 Next Steps for Data Infrastructure

While this project successfully built a static pipeline, the next logical evolution for this data architecture would include:

- **Automation:** Wrapping the Python script in a scheduler (like Apache Airflow) to automatically fetch daily CSVs and append them to the PostgreSQL database.
- **Predictive Analytics:** Utilizing the historical volume data to build a basic time-series forecasting model to predict inventory requirements for upcoming months, preventing stock-outs of high-performing "Sets".

End of Documentation

Prepared using Python, PostgreSQL, and Power BI.